

Tracked but Suppressed: How LLMs Fail State Tracking

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Abstract

Language models pretrained on trillions of tokens cannot reliably retrieve the current value of a variable that has been overwritten multiple times in their own input — a skill that underpins multi-turn dialogue, retrieval-augmented systems, and tool-augmented agents. Across 20 pretrained models (14 open-weight from 270M to 9B parameters, 6 proprietary frontier), this failure manifests as a general loss of position-indexed retrieval: consistently higher accuracy on the first value than on the current value (mean gap +27 percentage points, range +6 to +53 pp at representative cells), and near-zero accuracy on intermediate positions of the update stream. The failure is not architectural. A 7.4-million-parameter LoRA — 0.24% of one model’s weights — restores accuracy to 93–100% on harder held-out cells; a training-free format change produces similar recovery. Across 79 published checkpoints from two model families, the failure is present at the earliest pretraining checkpoint we can evaluate and survives every post-training stage. The capability is tracked in pretrained models but suppressed downstream — present in mid-layer representations, absent from the output. Mechanistically, LoRA amplifies an attention-head circuit that already attends partially to the latest value in the pretrained model and installs downstream redundancy that absorbs upstream perturbations.

1 Introduction

Language models are deployed in settings where the value of a variable changes over time and the model must use the current value, not an earlier one. A multi-turn assistant must respect the most recent user correction; a tool-augmented agent must read the current contents of a variable after a sequence of writes. Failures of this skill have been documented at the application level — Laban et al. (2025) report sharp degradation in multi-turn dialogues as state updates cascade — but the underlying mechanism, and whether the failure is intrinsic to current models, is not understood.

Figure 1 illustrates the task: K interleaved keys each receiving N updates, with the model asked for either the first (FVQ) or current (CVQ) value of a target key (formal setup in §3). Both queries are trivially solvable from context, so failure indicates a retrieval problem, not a length or memory limit.

We evaluate 20 pretrained models spanning 14 open-weight checkpoints (270M–9B parameters) and 6 proprietary frontier models. At a representative cell per model block,

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Stream ($K=3$, $N=3$, randomly interleaved):

color: red size: large shape: round
size: small color: blue shape: square
shape: triangle color: green size: medium

FVQ: What was the *first* value of color? → **red**

CVQ: What is the *current* value of color? → **green**

Figure 1: Task illustration ($K=3$, $N=3$). The stream randomly interleaves updates for all K keys. To answer CVQ correctly, a model must isolate color entries from competing keys and identify the most recent one among them.

FVQ–CVQ gaps range from +6 to +53 percentage points, with the majority exceeding +20 pp (Table 1, 12 models shown; Figure 2). The gap takes two shapes: **primacy** (FVQ exceeds CVQ; majority of cells) and **reversal** (CVQ exceeds FVQ but both below ceiling; smaller open-weight models at high load). When we ask for any intermediate position k of a target key ($1 < k < N$), baseline accuracy drops below 5% within two positions of the stream start (Figure 3; full sweep across six cells in Appendix F), indicating a general loss of position-indexed retrieval rather than a first-vs-current confusion.

The failure is not an architectural limit: a small 0.24%-parameter LoRA and a training-free format change both restore accuracy to 93–100% on harder held-out cells.

We make the following contributions:

- **Behavioural characterisation:** FVQ–CVQ gap across 20 models, including a previously undocumented reversal regime (§4).
- **Two recoveries:** task-specific LoRA and training-free format change both close the gap (§5).
- **Training-dynamics analysis** across 79 checkpoints from two model families; failure persists at every documented stage (§6).
- **Mechanism:** LoRA amplifies a partial v_{last} -attending circuit already present in pretraining and installs downstream redundancy; signatures replicate on Gemma-3-4b-it (§7).

2 Related Work

Retrieval failures in LLMs. Application-level work documents that LLMs struggle as in-context state changes: multi-turn dialogues degrade as state updates cascade (Laban et al. 2025); entity-state tracking is unevenly present across models (Kim and Schuster 2023; Kim, Schuster, and Toshniwal 2024; Rezaee, Camacho-Collados, and Pilehvar 2025); and most closely, Wang and Sun (2025) construct a 46-key stream with up to 400 updates and a single current-value query, reporting a log-linear accuracy decline they conclude is fundamental. The position-dependent failure parallels the static-context “lost in the middle” pattern (Liu et al. 2024), here arising under in-context updates rather than retrieval from a fixed document. Barbero et al. (2024) give a theoretical account of a related limit: signal propagation in decoder-only Transformers can map distinct input sequences to arbitrarily close final-token representations, which they show provably impairs counting and copying. Format sensitivity can shift accuracy by up to 76 percentage points (Sclar et al. 2024); our format intervention is a controlled instance. Knowledge-editing methods (Meng et al. 2022, 2023) address an adjacent failure — updating *parametric* facts stored in weights — whereas our setting requires tracking *in-context* updates without any weight change. This line of work characterises the failure; we show it is recoverable from the pre-trained architecture by either a 0.24%-parameter LoRA or a training-free format change, and trace the recovery to specific attention-head circuits.

Fine-tuning surfaces latent capacity. Recent work has shown that capabilities not visible in pretrained models can be surfaced with minimal supervision. Zhou et al. (2023) demonstrate that 1,000 curated dialogues are sufficient to teach a pretrained model to follow chat-style instructions, attributing this to pretraining having already done the work of knowledge acquisition. Sun and Dredze (2025) make a more general claim across 18 datasets: pretraining improves models in latent ways only visible after fine-tuning. We use standard low-rank adaptation (Hu et al. 2022; Aghajanyan, Gupta, and Zettlemoyer 2021). Our contribution is the concrete instance with mechanism: a capability pretrained LLMs systematically fail at, surfaced by an 18,000-example LoRA across held-out cells, datasets, and $\sim 10\times$ context lengths, and traced to upper-middle-layer attention heads.

Mechanism and pretraining dynamics. Mechanistic interpretability has identified circuits for specific capabilities (Olsson et al. 2022; Wang et al. 2023; Conmy et al. 2023; Li, Guo, and Andreas 2025), including how LMs bind entities to attributes and retrieve those bindings (Feng and Steinhart 2024; Gur-Arieh, Geva, and Geiger 2025); our task isolates failure of the positional retrieval path under destructive in-context updates. Barbero et al. (2025) analyse the first-token attention sink as a mechanism that limits over-mixing of representations across depth; we instead localise heads that route attention to the most recent value of a key. Closest on method, Mishra (2025) report that summarization fine-tuning shifts the function of 62% of attention heads; we apply the same approach to a mechanistic capability rather than a style task. On the pretraining-capability side, Wei et al. (2022)

and Schaeffer, Miranda, and Koyejo (2023) debate whether abilities emerge at scale; we ask the *non-emergence* question — this skill does not emerge at any scale we test. Allen-Zhu and Li (2024) parallel our claim for memorised knowledge: pretrained models store the information but do not develop a sufficient retrieval pathway by default. We add the mechanistic counterpart for an in-context capability: when the retrieval pathway is surfaced by a small fine-tune, it operates by amplifying a partial circuit already present in pretraining.

3 Task and Setup

Task. Let $\mathcal{K} = \{k_1, \dots, k_K\}$ be a set of keys and let $V_i = (v_{i,1}, \dots, v_{i,N})$ be an ordered update sequence for key k_i . The input stream σ is a random interleaving of all $K \times N$ pairs $\{(k_i, v_{i,j})\}$, subject to the constraint that no two consecutive entries share the same key. The model receives σ and a target key k_t and is asked one of three queries: **FVQ** (first-value, expected answer $v_{t,1}$); **CVQ** (current-value, expected answer $v_{t,N}$); or **IVQ** (intermediate-value, expected answer $v_{t,k}$ for some $1 < k < N$). Figure 1 shows a small instance. The intermediate-value query is used in the multi-model sweep (§4; Appendix E) and the dense-retrieval analysis (§5; Appendix F).

Datasets. **Arbitrary-Single (ARB)** uses a pool of 2,300 single-token English words across 46 semantic categories; single-token values keep cross-entropy and probing computations clean and make it the dataset for the mechanistic analysis (§7). **Semantic-Multi (SEM)** uses real category members as multi-token values across the same 46 categories (2,403 entries total) and is our primary behavioural dataset — semantic coherence between key and value limits template-matching shortcuts. Full construction details are in Appendix A.

Grid and trials. We evaluate models on a unified grid $K \in \{2, 3, 5, 7, 10, 15, 20, 25, 30\}$, $N \in \{5, 7, 10, 15, 20, 30, 50, 75, 100\}$, with up to 200 trials per (K, N) cell per query type. Cells where the prompt exceeds a model’s context limit or where the dataset pool cannot supply N distinct values per key are excluded (specifics in Appendix A).

Metric. For each (K, N) cell we report FVQ and CVQ accuracy as string-match proportions (matching rule in Appendix B) with Wilson 95% confidence intervals (Wilson 1927). The primary metric is the **gap** $\Delta = \text{FVQ accuracy} - \text{CVQ accuracy}$; we call a gap **substantial** when $|\Delta| > 0.05$. All inferences use greedy decoding (temperature 0) to eliminate sampling variance.

Models. We evaluate **11 open-weight models** from three major families — Qwen2.5 (0.5B–3B; instruct and base), Qwen3.5 (0.8B–9B; instruct), Gemma-3 (270M–4B; instruct) — and **6 proprietary frontier models** from OpenAI (GPT-4.1, GPT-4.1-mini), Anthropic (Claude-4.5-Haiku, Claude-4.5-Sonnet), and Google (Gemini-2.5-Flash, Gemini-2.5-Pro). Three additional smaller models — TinyLlama-1.1B, StableLM-2-1.6B, Pythia-410M — appear only in the appendix. The full model list is in Appendix C.

Model	FVQ	CVQ	Gap	% Ctx
<i>Proprietary models, $K=20, N=50$</i>				
GPT-4.1	1.00	0.71	+0.29	0.7%
GPT-4.1-mini	1.00	0.74	+0.26	0.7%
Claude-4.5-Haiku	1.00	0.53	+0.47	3.3%
Claude-4.5-Sonnet	1.00	0.74	+0.26	0.7%
Gemini-2.5-Flash	1.00	0.54	+0.46	0.6%
Gemini-2.5-Pro	0.90	0.77	+0.13	0.7%
<i>Open-weight models, $K=5, N=30$</i>				
Qwen2.5-3B-Instruct	0.61	0.55	+0.06	3.1%
Qwen3.5-2B	0.62	0.48	+0.14	0.4%
Qwen3.5-4B	0.86	0.64	+0.22	0.4%
Qwen3.5-9B	0.86	0.79	+0.07	0.4%
Gemma-3-1b-it	0.55	0.02	+0.53	3.1%
Gemma-3-4b-it	0.97	0.68	+0.29	0.8%

Table 1: FVQ and CVQ at the smallest cell per block where every model has gap $> +0.05$ ($K=20, N=50$ proprietary; $K=5, N=30$ open-weight). % Ctx is the prompt’s share of the model’s advertised context window. No model uses more than **3.3%** of its window at the cell it fails on.

4 Pretraining Fails at In-Context Update Retrieval

Every model we evaluate exhibits a substantial FVQ–CVQ gap. Table 1 reports, for each model block, the smallest cell at which every model’s gap exceeds $+0.05$; gaps there reach $+0.53$, and no model uses more than 3.3% of its context window. Per-cell evidence across the full (K, N) grid for all 20 models is in Appendix D. Figure 2 shows the trajectory in two open-weight models under three axis scans, including one that reaches the reversal regime.

Both shapes appear (Fig. 2). **Primacy** holds in every cell of Table 1. Under **reversal** CVQ exceeds FVQ but both sit well below ceiling, CVQ in the 50–65% range, in smaller open-weight models at high load. Neither is correct retrieval.

Intermediate-position queries ($1 < k < N$) collapse uniformly: at $K=10, N=50$ under the default Plain format, accuracy drops below 5% within two positions of the stream start across all five open-weight models and three of six proprietary models tested (Appendix E). Gemini-2.5-Pro is the only model at near-ceiling on this cell — but even it exhibits a $+0.13$ FVQ–CVQ gap at the harder $K=20, N=50$ cell (Table 1). The failure is universal; what differs is the operating point at which it becomes visible.

5 Two Recoveries: Format and LoRA

5.1 Format intervention

If the failure is about *locating* the relevant entries in the stream, making positions explicit at the surface should make the lookup easier. We test four formats: **Plain** (default, no markers); **Labeled** (per-entry index); **Block** (round headers); **Landmark** (round separators). Worked examples in App. B. Table 2 reports $K=10, N=50$ — the hardest cell in the format sweep — across 10 models (4 open-weight, 6 proprietary).

Block format closes almost all of the FVQ–CVQ gap on every open-weight model in Table 2: gaps of $+0.26$ to $+0.99$ under Plain fall to between -0.08 and $+0.10$ under Block. The recovery is genuine, not equalisation: under Block both queries reach 0.84 or above for every open-weight model, whereas under Plain CVQ falls to ≤ 0.23 in all four while FVQ stays well above it. Proprietary models follow the same pattern at smaller magnitudes; Gemini-2.5-Pro is the only model at ceiling on both queries under Plain, consistent with its non-collapse in §4.

Labeled and Landmark produce mixed recoveries (Table 2; per-model breakdown + 2×2 decomposition in App. E): round-grouping and numbering each contribute to CVQ recovery, but no single property alone restores both queries.

5.2 LoRA intervention

We fit a LoRA adapter (Hu et al. 2022) to Qwen2.5-3B-Instruct on 18,000 examples from a small training grid ($K \leq 10, N \leq 20$). The adapter adds 7.4M trainable parameters — 0.24% of model weights. We evaluate on a held-out grid extending to $K=30, N=75$, on held-out categories, and cross-distribution on Semantic-Multi (full configuration in Appendix G).

On the 28 held-out Arbitrary-Single cells, both FVQ and CVQ accuracy reach 93–100% post-adapter (gaps in Table 3; per-query accuracies in App. G), roughly $10\times$ beyond the training grid in stream length. The same adapter transfers to Semantic-Multi without modification (Table 4): at four held-out SEM cells, both queries reach 100% — including cells in the deep reversal regime with baseline gap up to -0.45 . The dense intermediate-position queries that collapsed under the base model in §4 also recover to near-ceiling at every queried position across six cells (Figure 3; Appendix F); simultaneous recovery across positions, not just the last, rules out a pure recency-amplification account.

Family control. The same recipe applied to Gemma-3-4b-it — a different model family with the opposite baseline regime ($\leq +87$ pp primacy vs Qwen’s reversal) — closes the gap on 28/28 held-out ARB cells (mean FVQ 99.4%, CVQ 97.0%) and 16/16 SEM OOD cells (mean FVQ 100%, CVQ 96.8%; Appendix G.1). The recovery is neither Qwen-specific nor regime-specific.

Negative control. The same recipe trained on GSM8K with identical hyperparameters learns arithmetic (16% $\rightarrow$ 70% GSM8K accuracy) but leaves the FVQ–CVQ gap unchanged on every held-out cell (Appendix G); the recovery is task-specific.

Two loci, one outcome. Format and LoRA produce the same behavioural outcome via *operationally distinct internal loci*: pooled $\Pr(\text{attend to } v_{\text{last}})$ on the 15 LoRA-promoter heads is 0.16 under Block versus 0.68 under LoRA, with 0 of 15 heads reaching LoRA-level activation (App. J). Both restore intermediate-position retrieval (App. E; Figure 3). Together they demonstrate the capability is *latent*: present in the pretrained architecture but not surfaced by default.¹ §6

¹We treat a capability as latent when (i) a small, task-matched fine-tune restores it and (ii) the mechanism amplifies pre-existing

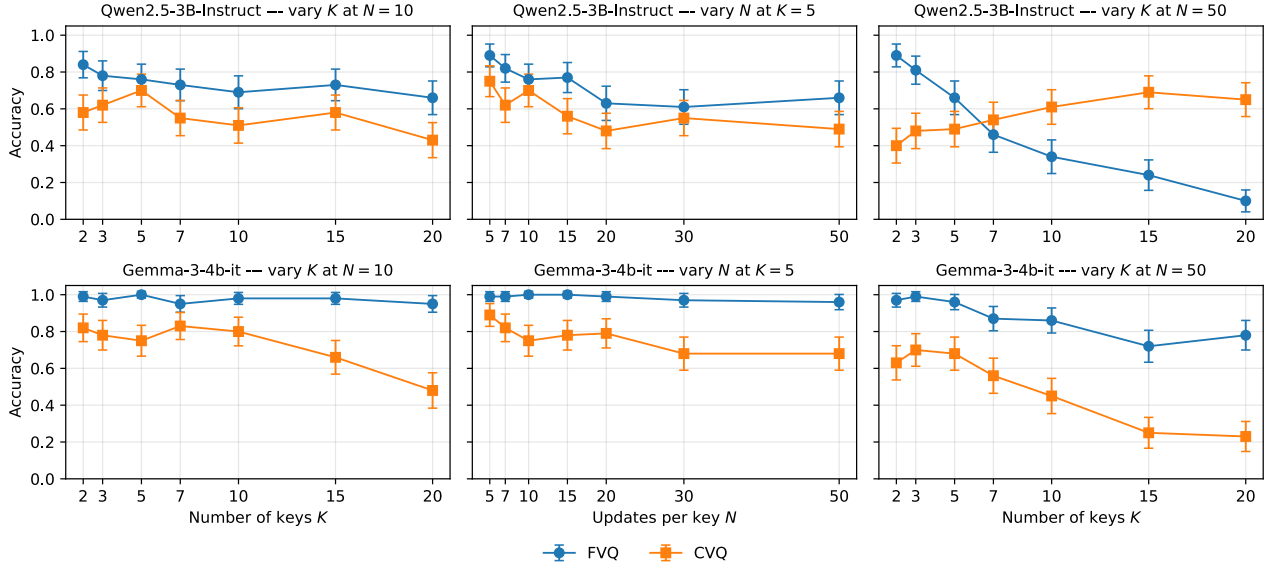


Figure 2: Qwen2.5-3B-Instruct (top) and Gemma-3-4b-it (bottom) across three axis scans on Semantic-Multi. Qwen crosses into the reversal regime at $K \approx 7$ (right column); Gemma stays in primacy throughout.

Model	Plain	Labeled	Block	Landmark
<i>Open-weight models</i>				
Qwen2.5-3B-Instruct	0.49 / 0.23	0.34 / 0.39	0.84 / 0.92	0.71 / 0.27
Qwen3.5-2B	0.95 / 0.01	0.85 / 0.93	1.00 / 0.98	0.80 / 0.07
Qwen3.5-4B	0.99 / 0.00	0.93 / 0.97	1.00 / 1.00	0.99 / 0.47
Gemma-3-4b-it	0.92 / 0.01	0.39 / 0.47	0.99 / 0.89	0.92 / 0.00
<i>Proprietary models</i>				
GPT-4.1	1.00 / 0.69	1.00 / 0.96	1.00 / 0.98	0.99 / 0.99
GPT-4.1-mini	1.00 / 0.77	1.00 / 0.94	0.99 / 0.99	0.78 / 1.00
Claude-4.5-Haiku	1.00 / 0.43	1.00 / 0.95	1.00 / 0.99	0.91 / 0.90
Claude-4.5-Sonnet	0.99 / 0.80	1.00 / 0.81	1.00 / 1.00	0.96 / 0.99
Gemini-2.5-Flash	1.00 / 0.76	1.00 / 0.98	1.00 / 1.00	1.00 / 0.95
Gemini-2.5-Pro	1.00 / 1.00	1.00 / 1.00	1.00 / 1.00	0.54 / 0.99

Table 2: Format-intervention results on Semantic-Multi ($K=10, N=50$). Each cell shows **FVQ / CVQ accuracy**; gap $\Delta = \text{FVQ} - \text{CVQ}$. Plain is the default condition from §4. Wilson half-widths in Appendix E.

asks whether any stage of training surfaces it without help; §7 asks what changes mechanistically when LoRA recovers it.

6 Does Any Training Stage Fix It?

We ask whether emergence is hiding at any training stage we have not yet evaluated. We track the FVQ–CVQ gap across 32 published SmolLM3-3B pretraining checkpoints (Hugging Face Smol Models Research Team 2025) spanning all three documented stages, plus its five post-training stages (mid-training, SFT, APO, LC-expert, final). Every checkpoint is evaluated on the same 32-cell

structure rather than installing it from scratch. The arithmetic LoRA control (App. G) fails (i); a model whose L30–L33 promoter heads had zero pre-LoRA $\text{Pr}(\text{attend to } v_{\text{last}})$ would fail (ii) and falsify the claim for this task.

(K, N) Arbitrary-Single grid, partitioned by stream-load (**low**: $K \leq 3$; **medium**: $K \geq 5, N < 20$; **high**: $K \geq 5, N \geq 20$); SmolLM2-1.7B (Ben Allal et al. 2025) contributes a further 42 checkpoints as a second family (App. H), for 79 in total.

Figure 4 answers the question: no. The gap is directionally biased at every checkpoint in every regime; no regime stabilises at zero. The five post-training stages do not close it; SmolLM2 shows the same picture across its four-stage curriculum.

We evaluate in completion format for cross-checkpoint comparability — the earliest pretraining checkpoints predate chat-template training. Re-running SmolLM3’s post-training stages under the model’s released chat template preserves the direction and ordering of the gap at every stage (Appendix H); the failure is not a completion-format artifact.

The same checkpoints are documented to succeed at multi-step reasoning, code synthesis, and long-context retrieval at

(K, N)	Base gap	+ LoRA gap
(7, 30)	+12%	+0%
(7, 50)	+6%	+4%
(10, 30)	−5%	+0%
(10, 50)	−8%	+0%
(15, 20)	−26%	+2%
(15, 30)	−25%	+2%
(15, 50)	−41%	+0%
(20, 30)	−30%	+2%
(20, 50)	−54%	+4%
(20, 75)	−50%	+4%
(25, 30)	−44%	+2%
(25, 50)	−38%	+2%
(25, 75)	−54%	+7%
(30, 30)	−48%	+2%
(30, 50)	−50%	+6%
(30, 75)	−55%	+6%

Table 3: Base vs +LoRA gap on 16 of the 28 held-out cells, extending to $K=30, N=75$ (adapter trained on $K \leq 10, N \leq 20$). Both queries reach 93–100% post-adapter; all 28 cells with FVQ/CVQ breakdown in App. G.

(K, N)	Base			+ LoRA (ARB-trained)		
	FVQ	CVQ	gap	FVQ	CVQ	gap
(7, 20)	60%	55%	+5%	100%	100%	+0%
(15, 20)	25%	55%	−30%	100%	100%	+0%
(15, 30)	35%	80%	−45%	100%	100%	+0%
(20, 30)	25%	65%	−40%	100%	100%	+0%

Table 4: Same LoRA, evaluated on Semantic-Multi (multi-token semantic values, never seen in training). All four held-out cells go from baseline reversal regime to gap 0%.

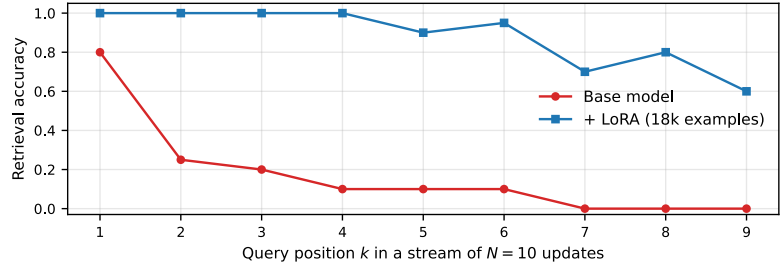


Figure 3: Per-position retrieval at $K=5, N=10$ on Arbitrary-Single, positions $k \in [1, N-1]$. Base collapses on intermediate positions; LoRA holds near ceiling at every position queried.

scales that exceed this task (Hugging Face Smol Models Research Team 2025; Ben Allal et al. 2025). The capability is recoverable by a small adapter (§5.2); pretraining does not surface it at any checkpoint we measure, motivating the circuit-level explanation in §7.

7 Mechanism: What LoRA Changes

The recovery in §5.2 shows the capability is in the pretrained architecture; §6 shows it remains unsurfaced through every documented training stage. We ask what changes inside the model when LoRA surfaces it. All measurements use Qwen2.5-3B-Instruct on Arbitrary-Single at $K=2, N=5$ unless noted (layer indexing 0-based, L0–L35; Table 5 for bootstrap 95% CIs, Figure 5 for per-layer trajectories and spatial localisation).

7.1 The substrate already exists in baseline

Three linear probes characterise what the pretrained model already represents. A condition-discrimination probe (FVQ vs CVQ prompt) reaches 100% accuracy from layer 6 onward in *both* models; LoRA adds nothing. A FVQ-correctness probe reaches 77–81% at L27–L33 in baseline — the model internally tracks its FVQ success. At the same layers a CVQ-correctness probe reaches only 57–65%, rising to 84–94% after LoRA (§7.2): whatever lets the baseline anticipate its FVQ success is largely absent for CVQ.

Logit lens shows the parallel pattern at the token level. At L32 of the baseline, $\Pr(v_{\text{last}})=0.34$ — the answer surfaces briefly — then decays to 0.25 at L35. *Found then suppressed*. At a harder cell ($K=2, N=50$) the answer barely surfaces (0.02 at L32, 0.11 at L35), tracking the behavioural failure. The readout machinery in layers 34–35 is intact (attention

patterns nearly identical pre/post-LoRA, Figure 5b); what the model lacks is propagation from the partial L32 answer to the final layer.

7.2 LoRA amplifies a partial v_{last} promoter circuit in layers 30–33

Attention routing locates the change. Fifteen heads in the L30–L33 band gain large v_{last} attention: pre-LoRA values of 0.05–0.23 rise to 0.60–0.86 post-LoRA — a roughly 4–12 $\times$ increase per head (Figure 5b). The largest shifts are L32H3 (0.11 $\rightarrow$ 0.78), L32H7 (0.13 $\rightarrow$ 0.78), L31H12 (0.17 $\rightarrow$ 0.81), and L31H15 (0.23 $\rightarrow$ 0.86). Layers 0–20 and 34–35 are essentially unchanged; the rewiring is spatially confined to a four-layer band.

The pre-LoRA values are not noise: the 15 promoters all sit above the 95th percentile of $\Pr(\text{attend to } v_{\text{last}})$ across all 576 baseline heads (median 0.001; fewer than 3% of heads exceed 0.05). §7.3 confirms causally that the heads carry v_{last} signal pre-LoRA: ablating the eight highest-attribution heads in the baseline drops L32 $\Pr(v_{\text{last}})$ by -0.28 (CI excludes zero).

The token-level consequence is immediate: post-LoRA L32 has $\Pr(v_{\text{last}})=0.99$ at $K=2, N=5$ and 0.62 at $K=2, N=50$, saturating to ~ 1.0 by L33–L35 at both. The representational consequence is parallel: the CVQ-correctness probe at L27–L33 rises from 57–65% to 84–94% ($\Delta=+0.36$, 95% CI $[+0.21, +0.50]$ at L33), installed in the same late layers that already carried the FVQ-correctness signal.

LoRA does not act by removing suppression: ablating the five baseline suppressor heads (L26H3, L27H3, L29H3, L29H4, L30H3) in the LoRA model releases v_{last} at L31 *more* than in baseline ($\Delta=+0.150$ vs $+0.078$). The suppres-

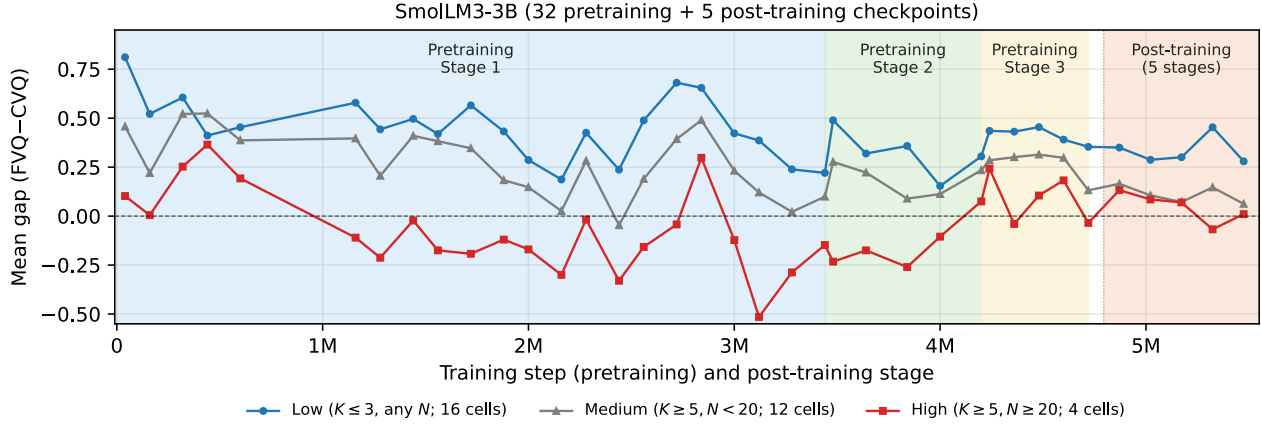


Figure 4: Mean FVQ-CVQ gap on Arbitrary-Single across the SmolLM3-3B (Hugging Face Smol Models Research Team 2025) training trajectory. Three load regimes: **Low** ($K \leq 3$, blue), **Medium** ($K \geq 5, N < 20$, grey), **High** ($K \geq 5, N \geq 20$, red). Pink band on the right: five post-training stages (mid-training, SFT, APO, LC-expert, final). No regime closes the gap at any documented checkpoint; SmolLM2-1.7B shows the same pattern (App. H).

sors keep firing post-LoRA; LoRA strengthens a parallel promoter pathway that overpowers them.

7.3 Two LoRA changes: amplification and downstream redundancy

To distinguish what LoRA *installs* from what it *amplifies*, we ablate the eight highest-attribution promoter heads in both models with paired trial seeds; the within-trial Δ s are directly comparable (Table 6). Paired differences below are reported as baseline - LoRA.

Two findings emerge. First, the heads are partial pre-LoRA carriers, and LoRA amplifies them: ablation drops $L32 \Pr(v_{\text{last}})$ by -0.28 in baseline (95% CI $[-0.34, -0.22]$, $n=200$) and by -0.40 in LoRA (95% CI $[-0.46, -0.33]$). The paired difference is $+0.12$ (95% CI $[+0.02, +0.21]$, excludes zero) — post-LoRA the same eight heads carry $\sim 40\%$ more causal effect at L32.

Second, propagation to the final layer differs. In baseline, the ablation leaves a small detectable residue at L35 ($\Delta = -0.05$, 95% CI $[-0.10, -0.00]$). In LoRA, the same ablation is absorbed by L35 ($\Delta = +0.00$, CI $[0, 0]$); paired difference -0.05 (CI $[-0.10, -0.00]$, excludes zero). LoRA has installed compensatory pathways at L33+ that re-stabilise $\Pr(v_{\text{last}})$ when the upstream signal is partially knocked out — redundancy that does not pre-exist in baseline.

Cross-family replication. Probing, logit-lens, and attention-routing signatures replicate on Gemma-3-4b-it (different family, opposite baseline regime). The CVQ-correctness probe at L26-L33 rises from 0.70-0.73 to 0.82-0.89 (Δ between $+0.11$ and $+0.19$, all eight layers with CI excluding zero); logit lens at L33 shows $\Pr(v_{\text{last}})$ rising from 0.50-0.80 to ≈ 1.00 across three cells; attention routing identifies a top-15 promoter cluster spread across L14-L32 ($\Delta +0.33$ to $+0.47$) — same late-layer functional pattern as Qwen, with a distributed rather than concentrated topology (App. I). Paired causal ablation on Gemma is left

to future work.

Synthesis. The four methods form a closed causal chain: heads in L30-L33 attend to v_{last} (routing) $\rightarrow v_{\text{last}}$ builds at L32 (logit lens) $\rightarrow$ a CVQ-correctness feature becomes linearly decodable (probing) $\rightarrow$ behaviour recovers. LoRA does two things in this chain: it amplifies a pre-existing partial promoter circuit (roughly $4-12\times$ in 15 heads of L30-L33) and installs compensatory pathways at L33+ that absorb upstream perturbations. The pretraining failure is a gain and robustness failure, not a representational one — §8 discusses what this implies.

8 Discussion

When each recovery is usable. Format intervention works at inference time but requires the deployer to control the prompt — agent harnesses with structured state, or instrumented multi-turn dialogue — and does not apply when prompt structure comes from user input or upstream components the deployer does not control. LoRA requires weight access, so it applies to open-weight or in-house fine-tuneable models but not to frontier inference behind a query interface. For frontier-only deployment paths neither recovery is available; our results predict the current-value retrieval failure persists.

Implications for pretraining. The next-token loss never requires overwriting an earlier binding: nothing in the objective rewards the current value over a valid earlier one. The 79-checkpoint sweep (§6) is what that predicts. A training signal targeting in-context update tracking is the natural next step.

9 Conclusion

Twenty pretrained models, from 270M to frontier scale, fail to report the current value of a variable they have watched

Method	Quantity	Baseline	+LoRA	Δ [95% CI]
Probe	CVQ-correctness, L33	0.58	0.94	+0.36 [+0.21, +0.50]
Probe	CVQ-correctness, L32	0.65	0.94	+0.29 [+0.22, +0.36]
Probe	CVQ-correctness, L30	0.57	0.84	+0.28 [+0.19, +0.37]
Probe	FVQ-correctness, L33	0.81	0.93	+0.12 [+0.05, +0.19]
Logit lens	$\Pr(v_{\text{last}})$, L32, $K=2$, $N=5$	0.34	0.99	+0.65 [+0.55, +0.73]
Logit lens	$\Pr(v_{\text{last}})$, L32, $K=2$, $N=10$	0.21	0.93	+0.73 [+0.63, +0.81]
Logit lens	$\Pr(v_{\text{last}})$, L32, $K=2$, $N=50$	0.02	0.62	+0.60 [+0.50, +0.69]
Logit lens	$\Pr(v_{\text{last}})$, L35, $K=2$, $N=50$	0.11	0.90	+0.79 [+0.71, +0.87]
Attention routing	$\Pr(\text{attend } v_{\text{last}})$, L32H3	0.11	0.78	+0.68 [+0.65, +0.71]
Attention routing	$\Pr(\text{attend } v_{\text{last}})$, L31H12	0.16	0.81	+0.66 [+0.60, +0.71]
Attention routing	$\Pr(\text{attend } v_{\text{last}})$, L32H7	0.12	0.78	+0.65 [+0.62, +0.68]
Attention routing	$\Pr(\text{attend } v_{\text{last}})$, L31H15	0.23	0.86	+0.64 [+0.57, +0.70]

Table 5: Bootstrap 95% CIs on key mechanistic Δ s at L30–L33, baseline vs +LoRA. Probing and attention-routing rows aggregate within the band; logit-lens rows give $\Pr(v_{\text{last}})$ at L32. Trial counts: probing 200 trials at $K=2$, $N=5$; logit lens 100 trials per cell; attention routing 50 trials at $K=2$, $N=30$. Bootstrap $n_{\text{boot}}=2000$, seed 17.

Ablation condition	Δ at L31	Δ at L32	Δ at L33	Δ at L35
Baseline + 5 baseline-suppressor heads	+0.078	−0.009	−0.058	+0.015
+LoRA + 5 baseline-suppressor heads	+0.150	−0.001	+0.000	+0.000
+LoRA + 8 LoRA-promoter heads	+0.007	−0.396	−0.005	+0.000
Baseline + 8 LoRA-promoter heads (control)	+0.014	−0.277	−0.007	−0.051

Table 6: Stage 3C four-way causal-ablation summary. Each row: paired within-run $\Delta = \text{ablated} - \text{normal}$ at four layers ($K=2$, $N=5$, CVQ; promoter rows $n=200$, suppressor rows $n=50$). **Bold row** is the baseline control — the eight LoRA-promoter heads carry a large L32 effect already in baseline, establishing them as partial v_{last} carriers pre-LoRA. The L33–L35 columns show LoRA’s new downstream redundancy: baseline leaks to the readout, LoRA absorbs it.

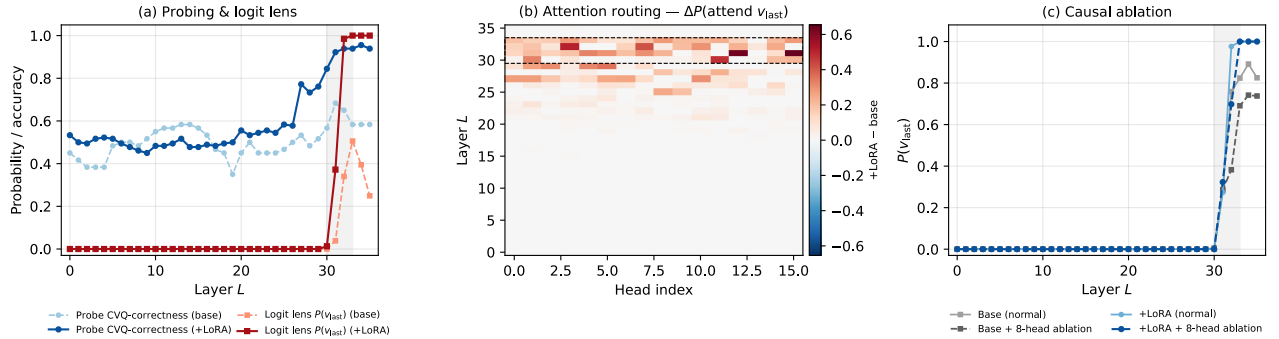


Figure 5: Mechanism summary for Qwen2.5-3B-Instruct ($K=2$, $N=5$). **(a)** Per-layer $\Pr(v_{\text{last}})$ (logit lens) and CVQ-correctness probe accuracy, baseline vs +LoRA — both signals converge on the L27–L33 band (shaded). **(b)** Per-head Δ in $\Pr(\text{attend } v_{\text{last}})$ across all 576 heads; the bright band at L30–L33 (dashed) contains the 15 promoter heads. **(c)** Stage 3C ablation of the eight top-attribution promoter heads drops $\Pr(v_{\text{last}})$ at L32 in *both* baseline and LoRA models; baseline ablation propagates to the readout, LoRA absorbs by L33+.

being overwritten — on prompts using under 4% of their context windows. The capability is nonetheless there: a 7.4M-parameter adapter lifts both first-value and current-value queries to 93–100%, and a format change recovers them with no weight update at all. The adapter amplifies attention heads that already carry the signal before it is trained, and across 79 checkpoints spanning two families, no stage of training surfaces it unaided. What pretraining fails to build here is not the representation but the path from it to the output.

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